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Machine Learning Based Water Saturation Prediction: Evaluating Ensemble and Neural Network Methods in Equinor's Volve Field

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This study compares machine learning (ML) methods – tree-based ensemble models and neural networks for predicting water saturation (Sw) directly from measured logs in Equinor's Volve field. The objective is to evaluate how ML-driven predictions can enhance traditional petrophysical workflows by providing rapid, accurate Sw estimations, supporting efficient reservoir management, and digital innovation in mature offshore oil fields. The value is to see how the water saturation front varies dynamically under field processes in a snapshot of time, offering insights that support timely decision-making and better control of production strategies.

Measured logs (Gamma Ray, Density, Neutron Porosity and Deep Resistivity) from 12 publicly available Volve wells were compiled alongside interpreted Sw logs. Data preparation involved automated log harmonization and outlier detection. Tree-based ensemble models (Light Gradient Boosting Machine (LightGBM), Random Forest, XGBoost) and neural network architectures (multi-layer perceptron) were trained and validated using subsets of wells. Models utilized only raw measured logs to ensure broad applicability. Performance was assessed via metrics (R^2 and RMSE), cross-plots, depth-wise comparisons, and petrophysical interpretability was evaluated through SHAP feature importance analysis using Python workflows.

Both modeling approaches effectively predicted water saturation (Sw), with ensemble-based techniques and neural networks showing comparable performance. A stacking regressor, composed of diverse base learners – including linear models, decision trees, ensemble regressors (Bagging, Extra Trees, Random Forest, CatBoost, XGBoost, LightGBM), KNN, SVR, and Elastic Net – with an LightGBM meta-learner, yielded high performance ($R^2 \sim 0.9445$, RMSE ~ 0.0895). The neural network achieved similar results ($R^2 \sim 0.9523$, RMSE ~ 0.0830). This stacked architecture leveraged the strengths of individual regressors, leading to improved generalization across lithological and fluid variation scenarios. Cross-plots showed strong alignment between predicted and actual Sw values, closely clustering along the 45° ideal line. Depth-wise log comparisons further confirmed accurate tracking of saturation transitions and reservoir heterogeneities. Feature importance analysis consistently identified resistivity and neutron-density logs as the most influential predictors—consistent with Archie's law and petrophysical intuition. The model was trained exclusively on raw logs, bypassing reliance on interpreted inputs such as porosity or shale volume,

thus enabling early-stage predictions before petrophysical analysis. Blind-well validation demonstrated strong generalization capabilities, while automated log harmonization ensured consistent input distributions, mitigating bias from manual normalization.

This study uniquely compares ensemble tree-based and neural network ML methods for direct S_w prediction from measured logs, removing dependence on interpreted parameters. The approach ensures broader applicability, consistency, and reduced bias in predictions, demonstrating significant potential for digital transformation and operational efficiency improvements in mature offshore oil fields.

Introduction

Reservoir management in mature North Sea fields presents unique challenges that require innovative approaches to maintain economic viability and optimize recovery [1]. The Volve field, like many mature fields in the region, faces declining production rates, complex water flooding patterns, and the need for cost-effective monitoring solutions [2]. These challenges are compounded by the heterogeneous nature of the reservoir formations and the varying quality of historical well log data accumulated over decades of field development [1,2].

Water saturation (S_w) stands as one of the most critical parameters in reservoir characterization, directly influencing hydrocarbon volume calculations, production forecasting, and field development decisions [3]. Accurate S_w determination is essential for identifying bypassed oil zones, monitoring water flood fronts, and optimizing enhanced oil recovery strategies [3]. Traditional methods for S_w calculation, primarily based on Archie's equation and its various modifications, have served the industry well but come with inherent limitations and uncertainties [1].

$$S_w^n = \frac{(a \times R_w)}{(\phi^m \times R_t)}$$

S_w Water Saturation Fraction of pore space filled with water <i>Range: 0 -1 (0% - 100%)</i>	R_t True Formation Resistivity Measured deep resistivity of the formation <i>Range: 0.1 – 10,000 $\Omega.m$</i>
R_w Formation Water Resistivity Resistivity of the water in pore spaces in cleaned rock <i>Range: ~ 0.01 – 1.0 $\Omega.m$</i>	ϕ Porosity Fraction of rock volume available for fluids <i>Range: 0.05 – 0.35 (5% - 35%)</i>
a Tortuosity Factor Accounts for pore network complexity <i>Typical: 0.6 – 1.2 (often ~ 1.0)</i>	m Cementation Exponent Related to rock consolidation and pore structure <i>Sandstone: 1.8 – 2.2, Carbonate: 2.0 – 2.5</i>
n Saturation Exponent Affected by wettability and pore geometry <i>Typical: 1.8 – 2.5 (often ~2.0)</i>	

Figure 1—Archie's Equation for Water Saturation Calculation

The classical Archie approach (Figure 1) relies on several assumptions and requires accurate determination of multiple parameters including formation water resistivity (R_w), cementation exponent (m), saturation exponent (n), and tortuosity factor (a) [1]. Each of these parameters introduces uncertainty into the final S_w calculation. For instance, the cementation exponent can vary significantly with lithology changes, from approximately 1.8-2.0 in clean sandstones to 2.2-2.5 in carbonates [3]. The saturation exponent is influenced by wettability conditions and can range from 1.8 to 2.5 or higher. Formation water resistivity variations due to salinity changes, particularly in water-flooded zones, add another layer of complexity [4].

Petrophysicists face several challenges during water saturation computation:

1. **Parameter Uncertainty:** The need to estimate or assume values for Archie parameters when core data is unavailable or limited
2. **Formation Heterogeneity:** Varying lithology and pore structures within the same reservoir interval requiring different parameter sets
3. **Clay Effects:** The presence of clay minerals affecting resistivity measurements and requiring more complex models like Waxman-Smiths or dual-water models [3]
4. **Fresh Water Effects:** In mature fields with extensive water flooding, mixing of injection and formation water creates variable R_w conditions
5. **Tool Vintage and Calibration:** Different logging tools and vintages introducing systematic biases in measurements
6. **Time Constraints:** The iterative nature of traditional interpretation workflows requiring significant manual effort

The rise of machine learning in petrophysics offers a paradigm shift in addressing these challenges [5,9]. Recent studies demonstrate that ensemble methods, particularly gradient boosting algorithms, have shown superior performance compared to traditional approaches, though their success often depends on the availability of pre-interpreted parameters like porosity and clay volume, which may not always be consistently available across different wells or fields. Additionally, while deep learning architecture has achieved impressive results in controlled settings, they frequently struggle with generalization when applied to geological environments that differ significantly from their training data [13]. Unlike traditional physics-based models that require explicit parameter definition, ML algorithms can learn complex, non-linear relationships directly from data. This data-driven approach has gained significant traction in recent years, with applications spanning lithofacies classification, porosity prediction, permeability estimation, and fluid saturation determination [9].

A review of recent literature reveals divergent experiences with ML applications in petrophysics. Some studies report exceptional success with R^2 values exceeding 0.95 for property predictions, while others caution about overfitting, limited generalizability, and the "black box" nature of certain algorithms [5,9]. The key differentiator often lies not in the choice of algorithm but in the quality of data preparation, feature engineering, and validation strategies employed.

What sets this study apart is the direct prediction of S_w from raw log measurements without intermediate interpreted parameters. Most published ML workflows for S_w prediction incorporate calculated properties like porosity, volume of shale, or water resistivity as input features [5]. While this approach can yield good results within a specific dataset, it perpetuates the uncertainties inherent in these intermediate calculations and limits the model's applicability to new wells where such interpretations may not be available or consistent.

The Volve field dataset, made publicly available by Equinor [4], provides an ideal testing ground for ML applications in petrophysics. With its comprehensive suite of wells, diverse geological settings within the Jurassic reservoir sequences, and expert-interpreted S_w curves for validation, it offers a realistic yet controlled environment for method development and comparison.

This study aims to demonstrate that machine learning can provide rapid, accurate, and consistent Sw predictions while maintaining petrophysical interpretability. By comparing ensemble tree-based methods with neural network approaches, we seek to identify the strengths and limitations of each, ultimately providing practical guidance for implementing ML-driven workflows in mature field operations.

Field and Data Description

Geological Context

The Volve field is in the Norwegian sector of the North Sea, approximately 200 kilometers west of Stavanger, in production license PL 046 at the southern end of the Norwegian Continental Shelf (Figure 2). Discovered in 1993 and produced from 2008 to 2016, Volve represents a typical North Sea oil field with complex reservoir architecture and production history [4].



Figure 2—Location map of the Volve Field in the Norwegian North Sea, showing its proximity to nearby oil fields and key cities

The field structure consists of an elongated four-way dip closure trending north-south, bounded by normal faults to the east and west. The main reservoir intervals are found within the Middle Jurassic Hugin Formation and the overlying Sleipner Formation, deposited in shallow marine to coastal environments during a major transgressive cycle [4].

Production History and Development Strategy

Volve was developed as a satellite field tied back to the Sleipner complex, utilizing a jack-up drilling rig and subsea production facilities. The field development strategy included:

- Initial production from two horizontal wells targeting the Hugin Formation
- Water injection support implemented early in field life to maintain reservoir pressure
- Later infill drilling to access bypassed oil and improve sweep efficiency
- Production of approximately 63 million barrels of oil over the field lifetime [4]

By the later stages of field life, water production had increased significantly, with water cuts exceeding 90% in some wells. This mature production profile, combined with the complex water flooding patterns, makes accurate water saturation determination crucial for identifying remaining oil opportunities.

Available Well Data

The dataset utilized in this study comprises logs from 12 wells made available through Equinor's data release initiative [4]. These wells provide comprehensive coverage of the field area and include various well types:

Training Wells (9 wells, numbered 0-8 in this paper):.

- 15/9-19: Discovery well with extensive coring and special log suites
- 15/9-F-1: First development well with complete log suite
- 15/9-F-1A: Sidetrack with updated logging technology
- 15/9-F-1B: Water injection well with time-lapse logging
- 15/9-F-1C: Producer with comprehensive formation evaluation
- 15/9-F-4: Horizontal producer through Hugin Formation
- 15/9-F-5: Deviated producer with behind-casing monitoring
- 15/9-F-10: Late-life infill well
- 15/9-F-11A: Producer in field periphery

Test Wells (3 wells, numbered 101, 103, 104 in this paper):.

- 15/9-F-11B: Sidetrack with modern logging suite
- 15/9-F-12: Infill producer in partially swept area
- 15/9-F-14: Peripheral well testing field limits

Log Suite and Coverage

The comprehensive logging program across Volve wells reflects evolving technology and operational objectives over the field life. The standard log suite includes:

Basic Logs and their typical measurement ranges (Available in all wells):.

- Gamma Ray (GR): 0-250 API units, measuring natural radioactivity
- Bulk Density (DEN): 1.5-3.0 g/cm³, from gamma-gamma density tools
- Neutron Porosity (NEU): -0.05-0.45 v/v, limestone units
- Deep Resistivity (RDEP): 0.1-10,000 ohm.m, various tool types

Extended Logs (Variable coverage):.

- Photoelectric Factor (PEF): For lithology identification
- Caliper (CALI): Borehole size for quality control
- Sonic logs (DTC, DTS): Compressional and shear slowness
- Medium/Shallow Resistivity (RMED): For invasion profiling
- Rate of Penetration (ROP): Drilling parameter

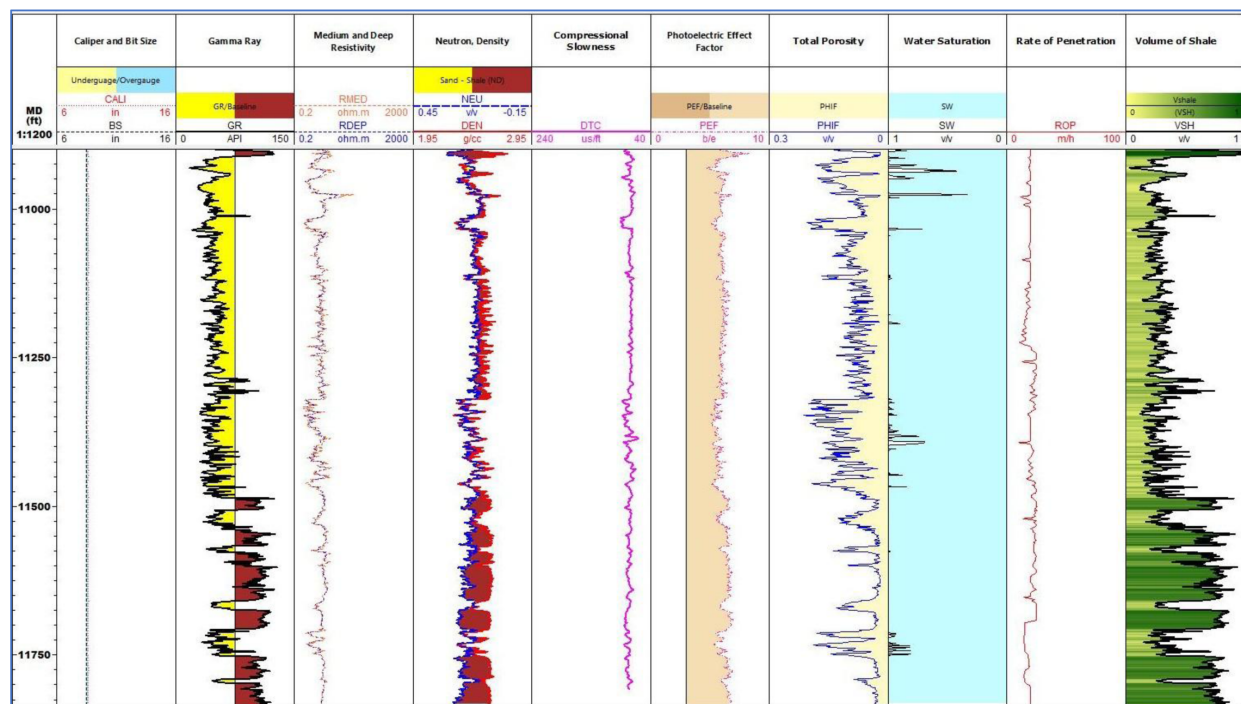


Figure 3—Comprehensive well log display from a Volve field well, showing the standard and extended log suite: gamma ray (GR), bulk density (DEN), neutron porosity (NEU), deep and medium resistivity (RDEP, RMED), photoelectric factor (PEF), caliper (CALI), compressional slowness (DTC), total porosity (PHIF), water saturation (Sw), rate of penetration (ROP), and calculated volume of shale (VSH). The image illustrates how technological advancements and operational goals influenced logging coverage across the field's development.

The variability in log coverage reflects different wells objectives, tool availability, and operational constraints. Early wells typically have more complete suites including sonic and photoelectric measurements, while later infill wells sometimes lack these specialized measurements. This heterogeneous data availability presents both challenges and opportunities for machine learning applications.

Data Quality Considerations

The Volve dataset exhibits several data quality characteristics typical of mature field operations:

1. **Tool Evolution:** Resistivity tools range from older laterolog devices to modern array induction tools, introducing potential calibration differences
2. **Borehole Conditions:** Variable hole conditions affect log quality, particularly in shaly intervals where washouts are common
3. **Depth Referencing:** Multiple drilling campaigns require careful depth matching between wells
4. **Environmental Corrections:** Different mud systems and borehole fluids require appropriate corrections
5. **Behind-Casing Logs:** Some intervals logged through casing with different tool physics

Petrophysical Interpretation Framework

The water saturation interpretations provided in the dataset were performed by experienced petrophysicists using industry-standard workflows. These interpretations incorporated:

- Core calibration from key wells
- Multi-well normalization to account for tool differences
- Variable Archie parameters based on rock typing

- Special core analysis (SCAL) for saturation exponent determination
- Integration with production data for validation [5]

This expert-interpreted S_w serves as the "ground truth" for training and validating machine learning models, though it's important to acknowledge that even expert interpretations contain inherent uncertainties and subjective decisions.

Data Preparation for Machine Learning (ML) model

The success of any machine learning application fundamentally depends on the quality and consistency of data input. In petrophysical applications, this challenge is amplified by the inherent variability in logging tools, borehole conditions, and measurement scales across different wells and time periods [7]. Our data preparation workflow was designed to address these challenges systematically while preserving the essential petrophysical information needed for accurate water saturation prediction.

Geological Correlation-Based Data Imputation

A critical challenge in the training dataset was the absence of water saturation interpretations in specific intervals across wells 1-7, particularly in a regionally extensive wet sand formation. This wet sand formation was clearly identified and interpreted in wells 0 and 8:

- **Well 0:** 12,544 - 14,120 ft (100% water saturation)
- **Well 8:** 9,807 - 12,098 ft (100% water saturation)

Through detailed well-to-well correlation, we identified a continuous water-bearing sand unit that could be traced across the field. Using geological correlation principles and the lateral continuity of this formation, we imputed $S_w = 1.0$ for the equivalent intervals in wells 1-7, which is shown in Figure 4, earlier missing in the input dataset.

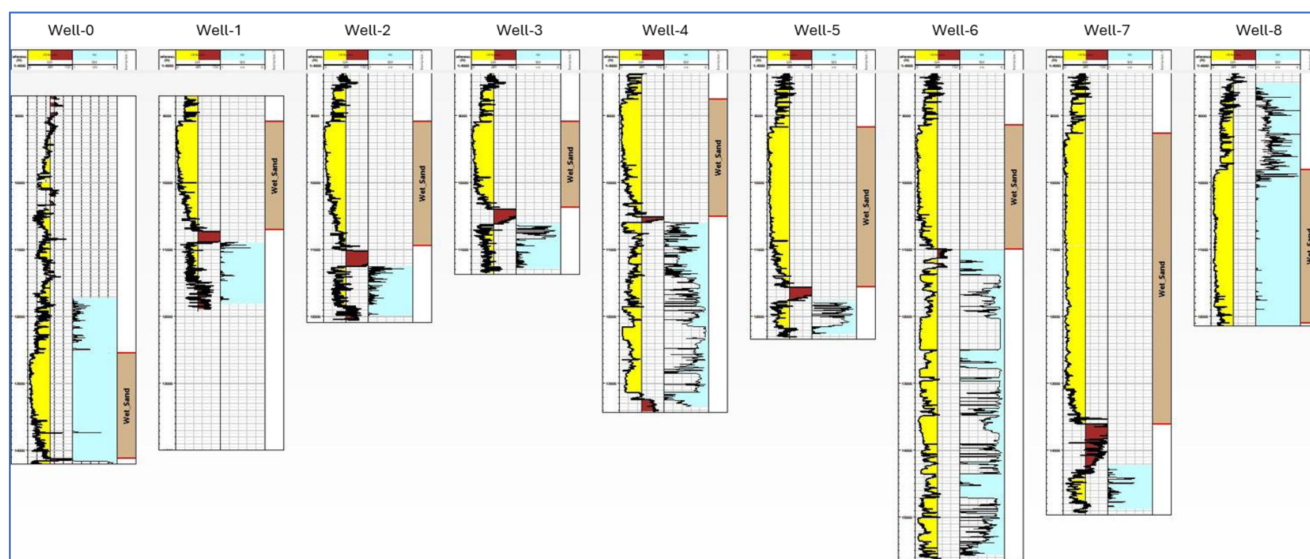


Figure 4—Well correlation panel showing the nine training wells (left: Well 0 to the right: Well 8) with the identified wet sand formation highlighted across the field, taking Well 8 as the reference well for stratigraphic analysis. The correlation demonstrates lateral continuity of this water-bearing interval.

Initial Data Assessment and Exploration

The raw dataset contained 318,967 data points across 17 features from 9 training wells were analyzed as shown in Figure 5. Initial exploration revealed several critical characteristics that guided our preprocessing strategy:

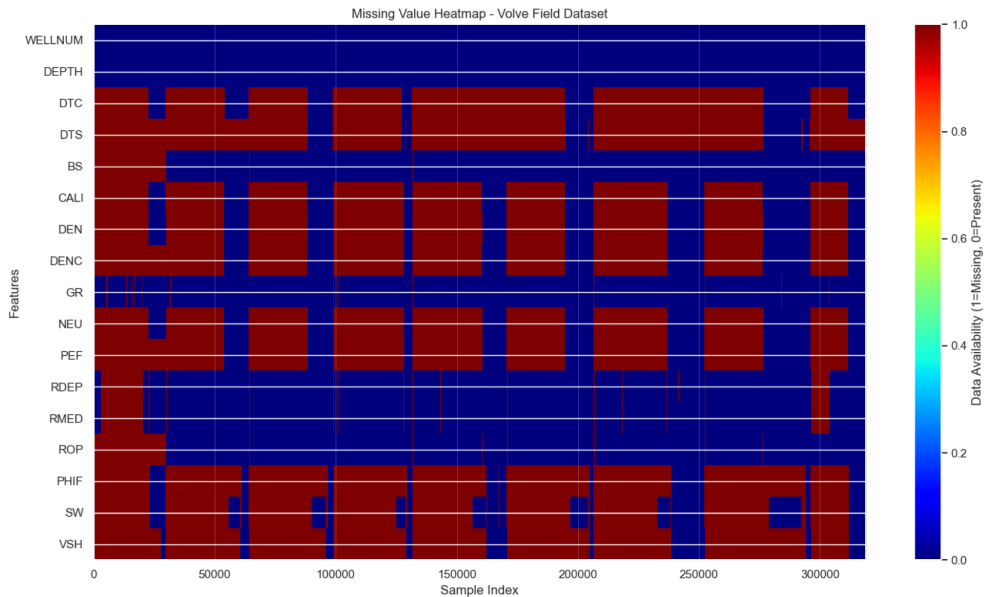


Figure 5—Missing value heatmap for the Volve field dataset using a jet colormap to visualize data availability across well log curves and depth intervals. Blue regions indicate available data, while red regions denote missing values. The color scale on the right quantifies data presence (0 = present, 1 = missing). This visualization highlights pronounced data gaps in specialized logs (DTS, VSH, PHIF) compared to more consistently recorded core measurements (GR, DEN, NEU, RDEP).

There was considerable variation in missing data types across the individual wells.

Total Records: 318,967		
Best Coverage GR (0.98%)		
Worst Coverage DTS (86.38%)		
Log Curve	Missing Count	Missing %
DTS	275,119	86.38%
VSH	273,867	85.86%
PHIF	271,653	85.17%
DTC	249,073	78.09%
PEF	230,431	72.24%
DENC	229,851	72.06%
SW	225,713	70.76%
NEU	223,528	70.08%
DEN	223,347	70.02%
CALI	222,810	69.85%
ROP	32,379	10.15%
BS	31,054	9.74%
RDEP	30,448	9.55%
RMED	30,214	9.47%
GR	3,119	0.98%

Figure 6—Quantitative summary of missing data percentages across all log curves in the Volve dataset (Total records: 318,967). Sonic logs (DTS: 86.38%) and interpreted parameters (VSH: 85.86%, PHIF: 85.17%) show the highest missingness, while basic log measurements (GR: 0.98%, RDEP: 9.55%) maintain nearly complete coverage. This structured missingness pattern reflects operational constraints, tool availability, and interpretation workflow variations across the field development history, guiding the selection of key logs (GR, RDEP, DEN, NEU) as primary features for machine learning model training.

The missing value patterns were not random but reflected operational decisions, tool availability, and interpretation coverage. This structured missingness required careful consideration in our preprocessing approach [10].

Log Selection Strategy

Given the variable data coverage and our objective of creating a robust, widely applicable model, we adopted a pragmatic approach to feature selection:

Primary Features Selected:

- Gamma Ray (GR): Universal availability, lithology indicator
- Deep Resistivity (RDEP): Formation conductivity indicator
- Bulk Density (DEN): Porosity and lithology information
- Neutron Porosity (NEU): Porosity and clay indication

This selection balances information content with data availability, ensuring the model could be applied to wells with basic log suites [3]. While additional logs like sonic or photoelectric factor could provide valuable information, their limited coverage would severely restrict the training dataset and model applicability.

Feature Engineering. Beyond the raw log measurements, we developed engineered features to capture known petrophysical relationships and improve model performance [5]. These features transform the raw measurements into physically meaningful parameters that enhance the machine learning models' ability to predict water saturation accurately.

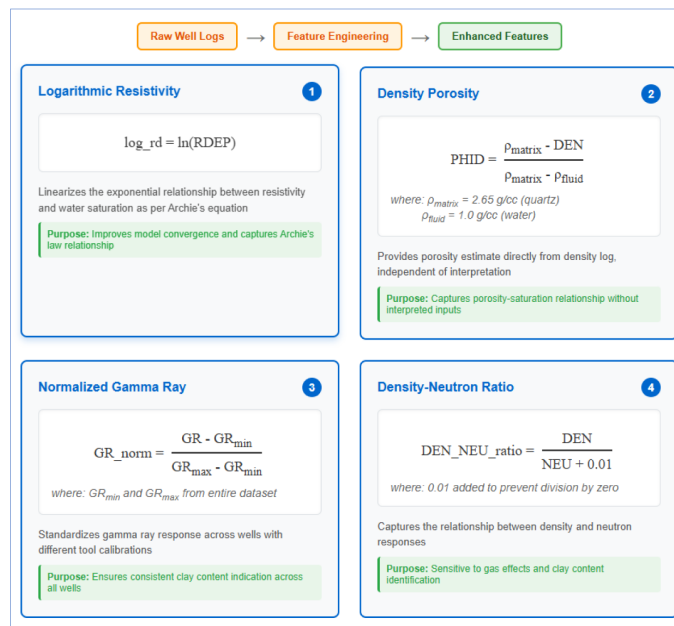


Figure 7—Engineered features transform raw well log measurements into physically meaningful parameters that can improve ML model performance

The engineered features include:

- **Logarithmic Resistivity (log_rd)**: Linearizes the exponential relationship between resistivity and water saturation, consistent with Archie's equation [1]. This transformation improves model convergence and explicitly captures the fundamental petrophysical relationship.

- **Density Porosity (PHID):** Provides a porosity estimate directly from the density log, independent of any interpretation [3]. Using standard matrix and fluid densities (2.65 g/cc for quartz and 1.0 g/cc for water), this feature captures the porosity-saturation relationship without relying on interpreted parameters.
- **Normalized Gamma Ray (GR_norm):** Standardizes gamma ray response across wells with different tool calibrations by scaling values between 0 and 1 using the dataset's minimum and maximum values. This ensures consistent clay content indication regardless of tool vintage or calibration differences.
- **Density-Neutron Ratio (DEN_NEU_ratio):** Captures the relationship between density and neutron responses, which is particularly sensitive to gas effects and clay content [3]. The small constant (0.01) prevents division by zero while maintaining the physical relationship.

These engineered features encode domain knowledge while maintaining independence from interpreted parameters, preserving our goal of prediction from raw measurements only. The integration of petrophysical principles through feature engineering significantly improved model performance, with feature importance analysis later confirming these engineered features among the top predictors of water saturation.

Feature Correlation Analysis and Descriptive Statistics

Correlation analysis using both Pearson and Spearman methods revealed distinct linear and non-linear relationships between well log features and water saturation (Sw). The Pearson correlation analysis (Figure 8) showed that PHID ($\phi = -0.75$), DEN ($\phi = 0.75$), and log_rd ($\phi = -0.55$) exhibited strong linear correlations with Sw, while GR_norm and GR demonstrated weak linear relationships ($\phi = -0.04$). However, Spearman correlation analysis (Figure 9) revealed a more nuanced picture, showing moderate negative monotonic relationships for GR and GR_norm ($\rho = -0.26$) with Sw.

This difference between correlation methods reflects the heterogeneous nature of the formations in the dataset. In clean formations (sandstones with minimal shale), water saturation is primarily controlled by porosity and resistivity measurements, making GR appear redundant for Sw prediction—hence the near-zero Pearson correlation. However, in shaly-sand formations, GR serves as a critical shale indicator that helps account for additional conductivity from clay-bound water, explaining the moderate Spearman correlation. In clean formations, gamma ray contribution is minimal for water saturation prediction, however, in shaly-sand formations, including GR in the model allows incorporation of the effect of extra conductivity from clay minerals.

Based on these insights, GR_norm was excluded in favor of the raw GR values, as subsequent feature scaling would handle normalization consistently across all features. The final feature set—log_rd, NEU, DEN, PHID, DEN_NEU_ratio, GR—was selected based on correlations with Sw, physical interpretability, and formation heterogeneity considerations. This set combines raw log measurements with engineered features that encode essential petrophysical principles while maintaining robustness across different formation types, from clean sands to shaly intervals.

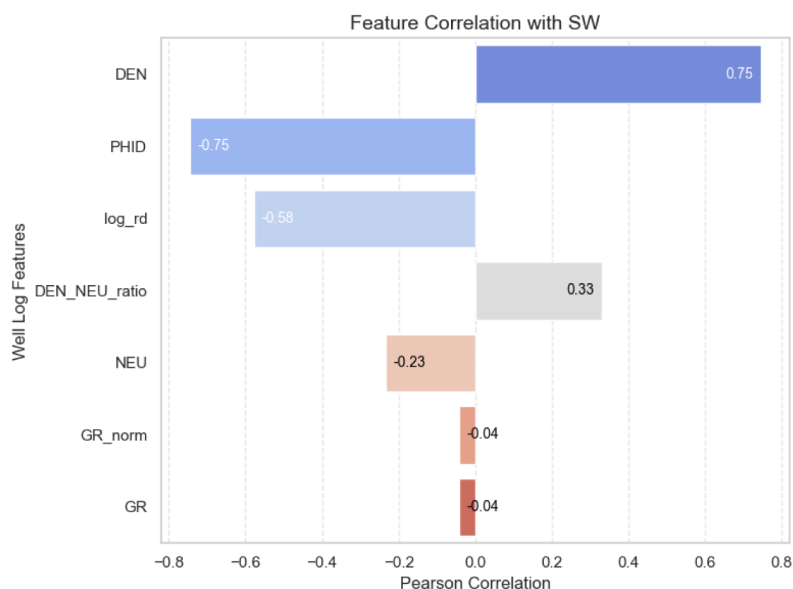


Figure 8—Bar plot showing Pearson correlation coefficients between well log features and water saturation (Sw)

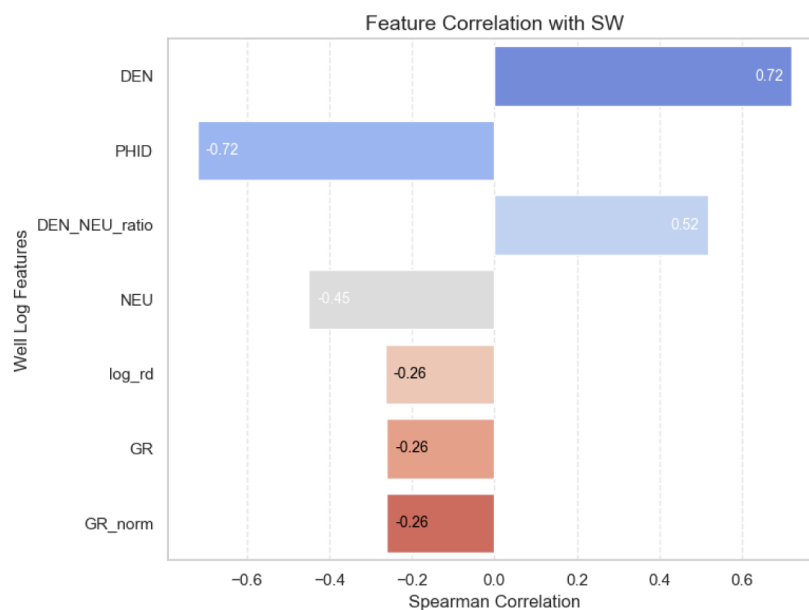


Figure 9—Bar plot showing Spearman correlation coefficients between well log features and water saturation (Sw)

Following feature selection and the removal of null rows and establishing a baseline cleaned dataset, initial data exploration revealed the presence of extreme outliers across multiple features. These outliers posed a serious risk to model stability and generalizability. The statistical summary of the cleaned input dataset (see [Table 1](#) below) highlighted unusually wide ranges and extreme maximum values that deviate significantly from the normal distribution of log measurements:

Table 1—Descriptive Statistics of cleaned dataset after doing feature selection and removal null rows without any missing value imputation

	GR	log_rd	NEU	DEN	PHID	DEN_NEU_ratio	SW
count	81704	81704	81704	81704	81704	81704	81704
mean	30.1	1.3	0.15	2.48	0.11	19.32	0.83
std	23.69	1.16	0.08	0.15	0.09	11.38	0.3
min	0.84	-2.04	0	1.63	-0.27	1.47	0.01
25%	10.83	0.65	0.1	2.38	0.04	11.94	0.78
50%	24.64	1.13	0.14	2.53	0.08	16.14	1
75%	43.56	1.82	0.19	2.58	0.16	23.79	1
max	304.33	11.29	1.46	3.09	0.62	405.73	1

Outlier Detection and Removal

Well log data often contains anomalous measurements arising from tool malfunctions, borehole conditions, or data transmission errors. These extreme values, if left unaddressed, can significantly distort machine learning model performance. To systematically identify and remove such anomalies, we employed the Isolation Forest algorithm [7].

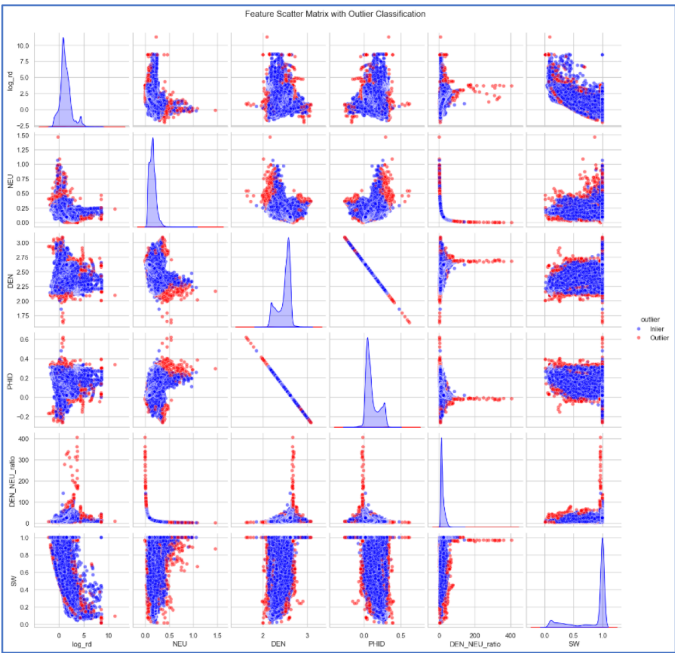


Figure 10—The pairwise scatter plots visualize relationships between input features and highlight observations classified as outliers (red) versus inliers (blue) by the Isolation Forest algorithm. Outliers typically cluster at the extreme edges of the feature distributions, confirming the presence of anomalous values such as negative porosity, extreme resistivity spikes, and unrealistic density-neutron ratios.

The Isolation Forest algorithm works by isolating observations through random feature selection and split value determination. Anomalies, being few and substantially different from most of the data, are isolated earlier in the tree structure [7].

We configured the algorithm with the following parameters:

- **Contamination factor:** 0.01 (assuming approximately 1% of the data to be anomalous)

- **Random state:** 21 (for reproducibility)
- **Number of estimators:** 100 (default setting)

Post-outlier removal results based on the cleaned dataset:

- **Total samples before outlier removal:** 81704
- **Inlier samples remained:** 80,886
- **Outlier samples removed:** 818
- **Actual outlier percentage:** ~1.0%

The scatter matrix shown in [Figure 10](#) clearly illustrates how the detected outliers (red points) are typically concentrated at the distribution boundaries or occupy physically implausible regions of the feature space. The small proportion of removed outliers primarily consisted of:

- **Physically impossible values** (e.g., negative bulk density or porosity)
- **Extreme resistivity spikes** typically associated with poor borehole conditions or logging tool errors
- **Abnormal neutron-density separations** indicating hole quality issues

By removing these extreme and erroneous values, we significantly improved the overall data quality and consistency. This step was critical to ensure better model convergence, more stable learning patterns, and improved generalization on unseen wells in subsequent modeling stages.

Train-Test Split Strategy

To ensure robust model validation and realistic performance assessment, we implemented a hierarchical data splitting approach that combines both random sampling and well-based holdout strategies:

Primary Split - Training Wells (Wells 0-8):

The nine training wells containing 80,886 samples after preprocessing were split using an 80/20 random sampling approach:

- Training Set: 64,708 samples (80%) used for model training
- Validation Set: 16,178 samples (20%) used for hyperparameter tuning and early stopping

This random split within the training wells ensures that the model sees diverse geological conditions during training while maintaining a held-out validation set for model selection.

Secondary Split - Blind Test Wells (Wells 101, 103, 104):

Three complete wells containing 8,212 samples were withheld entirely from the training process to serve as a blind test set. This well-based holdout approach provides the most rigorous test of model generalization, as these well:

- Representing different drilling campaigns and time periods
- Contain potentially unique geological features not seen in training
- Simulate real-world deployment where models must predict on entirely new wells

Feature Scaling – Standardization of Input Variables:

To ensure uniform feature contribution and numerical stability across machine learning algorithms, all input features were standardized using z-score normalization via StandardScaler [10] from Scikit-learn. This preprocessing step is especially crucial for distance-based algorithms (e.g., k-NN, SVM) and gradient-based optimizers used in neural networks.

Machine Learning Methodology

The development of robust machine learning models for water saturation prediction requires careful consideration of algorithm selection, hyperparameter optimization, and validation strategies. Our methodology employed both ensemble tree-based methods and neural network architectures to leverage their complementary strengths while maintaining petrophysical interpretability [5,6]. Figure 10 illustrates the complete machine learning workflow implemented in this study, from initial data input through model deployment.

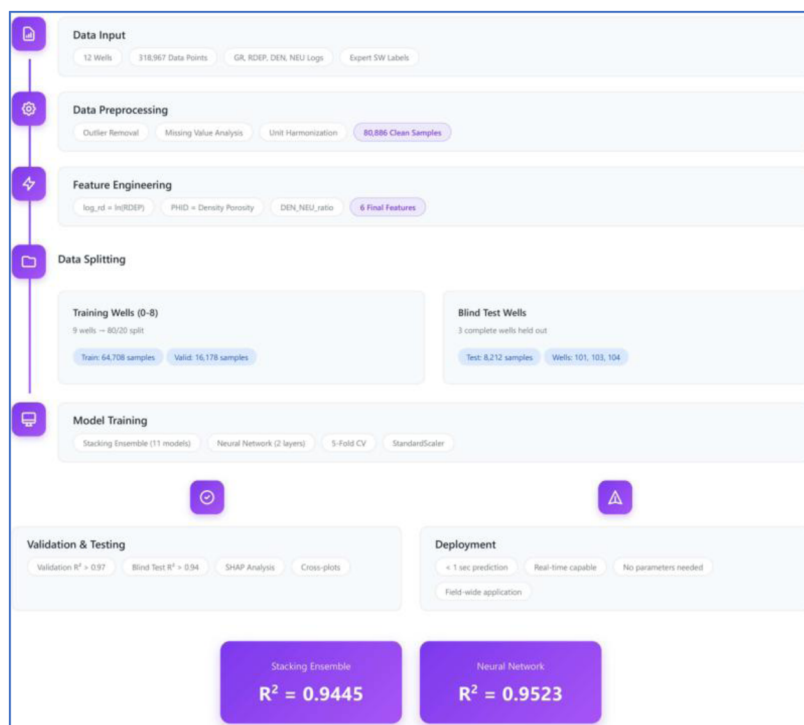


Figure 11—Machine learning workflow pipeline showing the systematic progression from data input to deployment.

The workflow encompasses seven key stages: (1) Data input from 12 wells with 318,967 data points, (2) Data preprocessing including outlier removal and harmonization resulting in 80,866 clean samples, (3) Feature engineering to create five physically meaningful features, (4) Hierarchical data splitting with 9 training wells (80/20 train-validation split) and 3 blind test wells, (5) Parallel model training using stacking ensemble and neural network architectures, (6) Comprehensive validation and testing, and (7) Deployment-ready models. Final blind test performance shows $R^2 = 0.9445$ for the stacking ensemble and $R^2 = 0.9523$ for the neural network.

Model Selection and Implementation

Our algorithm selection was guided by petrophysical considerations: the non-linear relationship between logs and water saturation [1], complex feature interactions in zones with gas or clay effects, the need for interpretable results [8], and robust handling of measurement uncertainties.

- **Linear Models:** Linear Regression and ElasticNet with L1/L2 regularization
- **Tree-Based Models:** Decision Tree, Random Forest, Extra Trees, and gradient boosting variants (XGBoost [2], LightGBM [6], CatBoost)

- **Other Methods:** K-Nearest Neighbors, Support Vector Regression, and Bagging Regressor

All implementations utilized scikit-learn [10] and respective specialized libraries.

Stacking Ensemble Architecture. The stacking approach leverages model diversity by combining predictions from 11 base models through a meta-learner [5]. Our architecture uses LightGBM as the meta-learner due to its efficiency in capturing non-linear combinations of base predictions [6]. The ensemble employs 5-fold cross-validation to generate out-of-fold predictions, preventing overfitting while ensuring the meta-learner trains on unseen data (Figure 12).

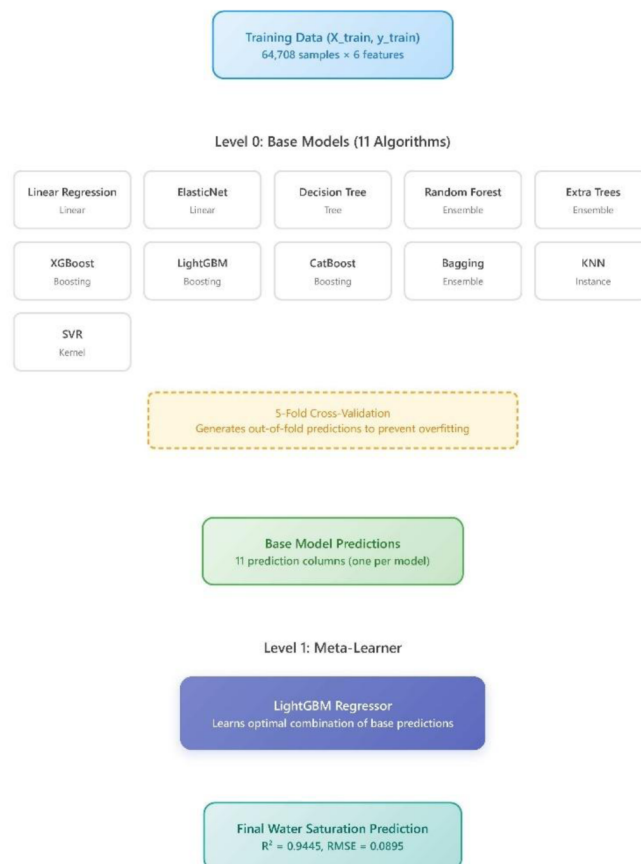


Figure 12—Stacking ensemble architecture showing the two-level hierarchy. Level 0 consists of 11 diverse base models spanning linear, tree-based, and kernel methods. These models are trained using 5-fold cross-validation to generate out-of-fold predictions, which serve as input features for the Level 1 meta-learner (LightGBM). The meta-learner learns the optimal combination of base model predictions to produce the final water saturation estimate.

This approach reduces error correlation between models, provides robustness against individual model failures, and typically achieves superior performance compared to single algorithms. The diversity of base learners—from simple linear models to complex gradient boosting machines—ensures comprehensive capture of different data patterns and relationships.

Neural Network Architecture. As an alternative to tree-based ensembles, we implemented a feedforward neural network (FFNN) to explore deep learning approaches. Our FFNN comprises of:

- Input layer: 6 features
- Two hidden layers: 64 neurons each with ReLU activation and 20% dropout
- Output layer: Single neuron with linear activation

Training utilized Adam optimizer (learning rate=0.001), MSE loss, and early stopping (patience=10) to prevent overfitting. Figure 13 shows the training history, demonstrating rapid convergence within the first 10 epochs and stable performance thereafter.

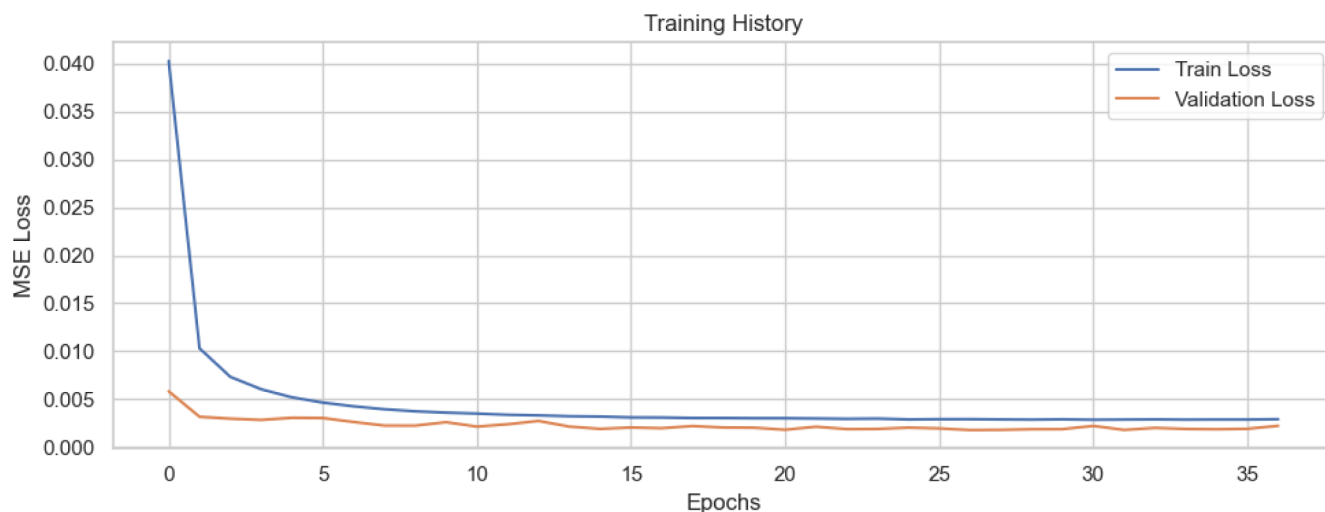


Figure 13—Neural network training history over 40 epochs. The model shows rapid initial convergence with training loss decreasing from 0.041 to below 0.005 within 10 epochs. The close alignment between training and validation loss curves indicates effective regularization through dropout layers, with no signs of overfitting. Early stopping with patience=10 ensured optimal model selection at the point of minimum validation loss.

Model Evaluation

Model performance was assessed using two primary metrics:

- **R² Score:** This metric quantifies the proportion of variance in the observed water saturation values that is captured by the model. In our study, R² scores consistently surpassed 0.94, demonstrating the model's robustness in learning from complex feature interactions and generalizing across varying lithologies and reservoir characteristics.
- **Root Mean Square Error (RMSE):** RMSE measures the average magnitude of prediction errors in the same unit as the target variable—water saturation. An RMSE below 0.10 reflects high fidelity in the model's predictions, with minimal deviation from actual measurements.

Both metrics were computed on the validation subset during model development and on a set of blind test wells for final evaluation. The consistently high R² scores and low RMSE values across different wells and formations confirm the model's effectiveness in capturing the non-linear dependencies inherent in petrophysical data and its applicability to diverse geological scenarios.

Feature Importance Analysis

Understanding feature contributions is crucial for model validation and petrophysical interpretation. We employed SHAP (SHapley Additive exPlanations) analysis [8]. The feature importance analysis was conducted on a sophisticated stacking ensemble model comprising eleven diverse base learners (Figure 14). And the SHAP analysis was specifically performed on the LightGBM meta-learner component of the stacking ensemble, providing insights into how the final model combines and weighs different features.

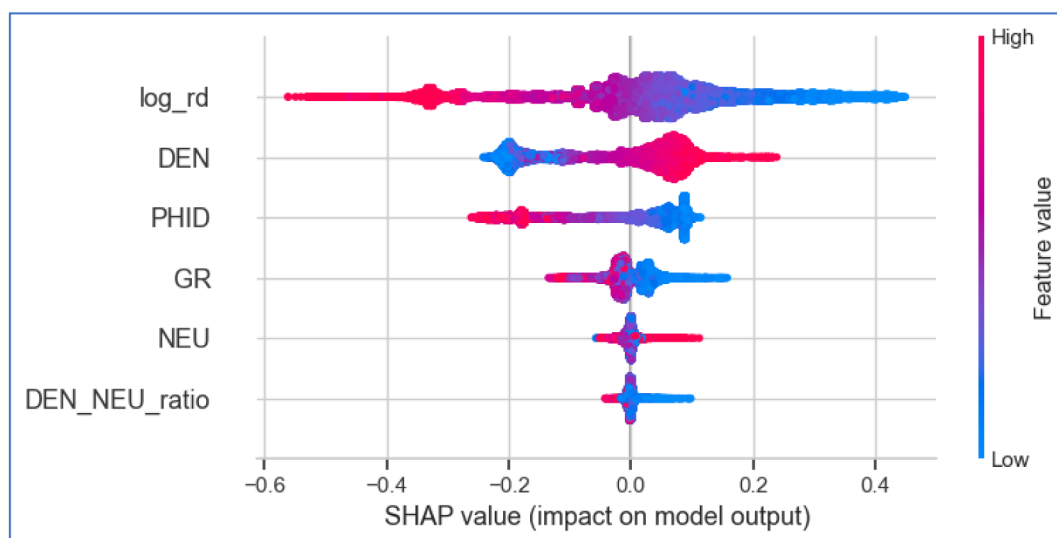


Figure 14—SHAP summary plot from the meta-learner showing feature importance and directional impact on water saturation predictions. Resistivity ($\log_{rd}$) demonstrates dominant influence with predominantly negative correlation, followed by density (DEN) and engineered porosity features. Gamma ray (GR) shows moderate importance with bidirectional contributions.

Key Findings:

- **Resistivity ($\log_{rd}$):** Dominant feature with widest impact range (-0.6 to +0.4 SHAP values). Predominantly negative correlation with water saturation aligns with Archie's law principles.
- **Density (DEN):** Secondary importance with SHAP values from -0.25 to +0.2, showing both positive and negative contributions reflecting complex density-saturation relationships.
- **Density Porosity (PHID):** Engineered feature with concentrated impact around zero and notable negative contributions, capturing inverse porosity-saturation relationships.
- **Gamma Ray (GR):** Moderate importance with balanced positive and negative SHAP values (-0.1 to +0.1), capturing shale volume effects on water saturation. Higher GR values (indicating the shalier intervals) show both increasing and decreasing impacts on water saturation, reflecting complex clay-bound water relationships and presence of kerogen in the source rock.
- **Neutron (NEU):** Moderate importance with tightly clustered SHAP values near zero, indicating consistent secondary contributions.
- **Density-Neutron Ratio (DEN_NEU_ratio):** Smallest impact range with subtle positive influences.

Petrophysical Validation: These importance rankings align with established petrophysical theory:

- **Resistivity Dominance:** Confirms Archie's equation as foundational for saturation determination
- **Porosity Controls:** Density and neutron responses capture pore space characteristics and lithology
- **Lithology Indicator:** Gamma ray incorporation accounts for clay content effects on water saturation, crucial for shaly sand formations
- **Feature Engineering:** Engineered features demonstrate meaningful contributions, validating domain expertise

This meta-learner SHAP analysis validates the model's petrophysical learning and demonstrates how the ensemble synthesizes diverse modeling perspectives for superior water saturation prediction.

Blind Well Validation Results

The ultimate test of model generalization comes from blind well prediction performance across three independent test wells that were completely withheld from the training process. Figures 15 – 18 demonstrates the comprehensive validation results for Wells 101, 103, and 104, showcasing both log-by-log predictions and statistical performance metrics.

Well-101 (Figure 15): Both models demonstrated excellent performance in this structurally complex well. The stacking ensemble achieved $R^2 = 0.960$ while the neural network achieved $R^2 = 0.968$, with both models accurately capturing the water saturation variations across multiple reservoir zones. The crossplot analysis reveals tight clustering around the 1:1 line with minimal scatter, particularly in the mid-range saturation values (0.3-0.7) that are critical for reserves estimation. Both models successfully identified the low saturation pay zones ($S_w < 0.4$) in the upper reservoir section while accurately predicting higher water saturations in the transition zones.

Well-103 (Figure 16): This well presented more challenging conditions with complex lithological variations, as evidenced by the highly variable gamma ray response. The neural network ($R^2 = 0.938$) slightly outperformed the stacking model ($R^2 = 0.925$) in this challenging environment. The log displays reveal excellent agreement between predicted and actual water saturations across the entire logged interval, with both models successfully handling the clay-rich intervals that typically challenge traditional Archie-based calculations. The crossplot demonstrates robust performance across the full saturation range, with slightly more scatter in the high-saturation regions ($S_w > 0.8$) where clay effects are most pronounced.

Well-104 (Figure 17): Representing the best individual well performance, both models achieved exceptional accuracy with the neural network achieving outstanding performance ($R^2 = 0.974$) compared to the stacking ensemble ($R^2 = 0.959$). This well's cleaner sandstone character, as indicated by the lower gamma ray values, provided optimal conditions for both modeling approaches. The log tracks show virtually identical predictions between the two methods, with both accurately capturing the sharp saturation contrasts at reservoir boundaries and the gradual transitions within flow units.

Figure 18 presents the combined crossplot analysis across all three blind wells, providing a comprehensive view of model performance. The stacking ensemble achieved an overall $R^2 = 0.9445$, while the neural network achieved $R^2 = 0.9523$. This slight advantage for the stacking approach likely reflects its ensemble nature, which provides more robust predictions by combining multiple modeling perspectives. Both crossplots demonstrate:

- **Excellent Linear Correlation:** Dense clustering along the 1:1 diagonal confirms unbiased predictions across the full saturation range
- **Robust Low-Saturation Prediction:** Critical for hydrocarbon reserves, both models accurately predict $S_w < 0.5$ with minimal scatter
- **Consistent High-Saturation Performance:** Even in challenging water-bearing zones ($S_w > 0.8$), both models maintain reasonable accuracy

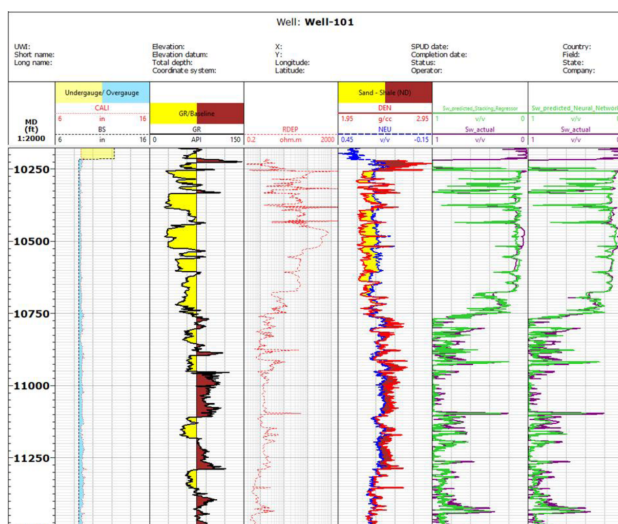


Figure 15—Well log display and crossplot analysis for blind Well-101 showing stacking ensemble ($R^2 = 0.960$) and neural network ($R^2 = 0.968$) water saturation predictions compared to actual interpretations.

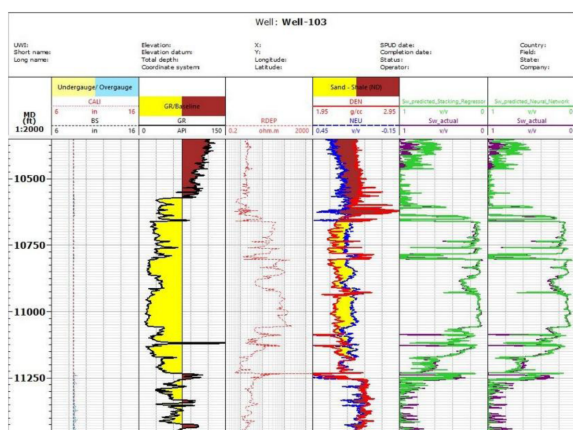
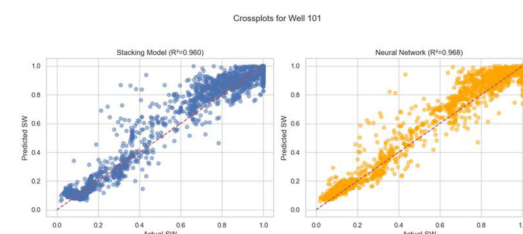


Figure 16—Well log display and crossplot analysis for blind Well-103 demonstrating stacking ensemble ($R^2 = 0.925$) and neural network ($R^2 = 0.938$) performance in geologically complex conditions with significant clay content.

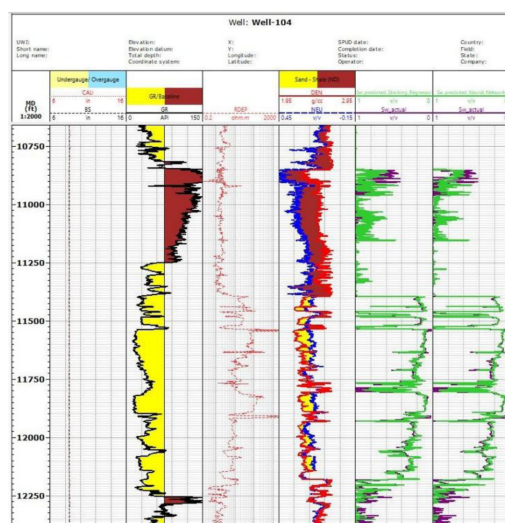
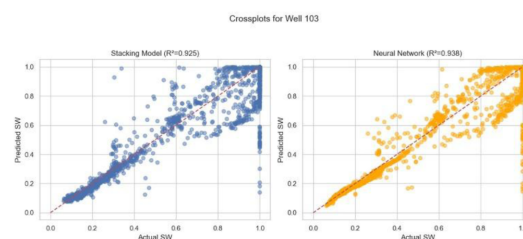
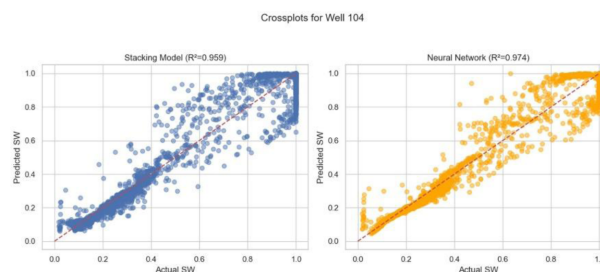


Figure 17—Well log display and crossplot analysis for blind Well-104 showing exceptional performance with neural network ($R^2 = 0.959$) and stacking ensemble ($R^2 = 0.974$) predictions in clean sandstone reservoir conditions.



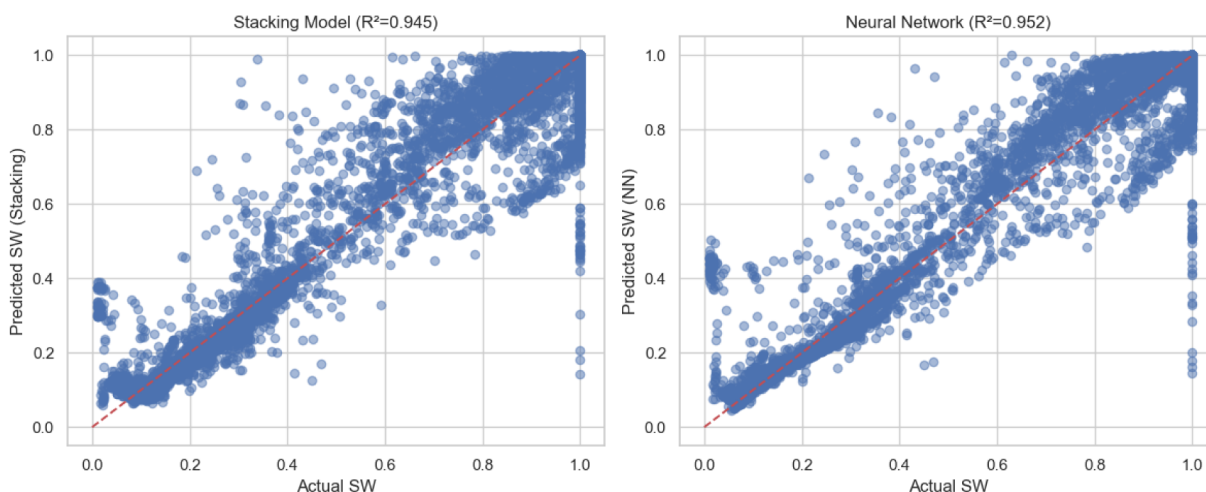


Figure 18—Aggregate crossplot analysis from all three blind wells showing overall stacking ensemble ($R^2 = 0.945$) and neural network ($R^2 = 0.952$) performance across diverse geological conditions.

Analysis and Interpretation

The comparable performance between stacking ensemble ($R^2 = 0.9445$) and neural network ($R^2 = 0.9523$) approaches indicates that algorithm architecture choice is less critical than data preparation and feature engineering for water saturation prediction. Both methods successfully captured non-linear log-saturation relationships, achieving accuracy matching traditional petrophysical interpretation. The convergence suggests we may be approaching theoretical prediction limits given inherent uncertainties in expert-interpreted training labels. The achieved RMSE of ~ 0.08 saturation units likely represent combined model error and training data inconsistencies.

Machine learning approaches offer distinct advantages over conventional Archie-based calculations. Parameter uncertainty elimination represents a significant benefit, as ML models implicitly learn cementation exponent (m), saturation exponent (n), and formation water resistivity (R_w) relationships, eliminating subjective parameter selection and error propagation that plague traditional methods. Automatic clay effect handling provides another major advantage, as complex clay corrections typically requiring sophisticated models like Waxman-Smits or dual-water [11, 12] approaches are inherently captured without explicit clay volume calculations or capacity exchange estimates.

Limitations and Considerations

Despite strong performance, several limitations must be acknowledged:

- **Training Data Dependency:** Model accuracy is inherently limited by the quality and coverage of training data. Wells with lithologies, fluid types, or log responses not represented in the training set may yield less reliable predictions [5]. The Volve dataset, while comprehensive, represents a specific geological setting that may not fully transfer to other fields.
- **Black Box Concerns:** While SHAP analysis provides insights into feature importance [8], the detailed decision-making process within ensemble trees or neural networks remains opaque. This can create challenges for regulatory approval or when detailed explanations are required for specific predictions.
- **Extrapolation Risks:** ML models generally perform poorly when extrapolating beyond their training domain [9]. Extreme log values, unusual geological features, or new logging tools may produce unreliable results without warning.

- **Quality Control Requirements:** Automated predictions still require expert oversight to identify geologically unreasonable results or artifacts from poor data quality. ML should augment, not replace, petrophysical expertise.
- **Applicability to Mature Fields:** Caution is warranted when applying these models to mature fields. Over time, both rock and fluid properties can evolve significantly from their original states (both salinity and saturation (S_w)). Since the models rely on raw log measurements, such subsurface changes may not be adequately captured. Consequently, predictions in mature reservoirs could be unreliable unless augmented with additional data sources, such as measurements behind casing or production history, to account for altered conditions. This challenge underscores the need for rethinking model applicability and validation strategies in aging assets.

Conclusion

This study demonstrates that machine learning, specifically stacking ensembles and neural networks, offers a highly effective approach for predicting water saturation directly from basic well logs. Both models achieved excellent performance on blind validation sets ($R^2 > 0.9$, $RMSE < 0.10$), consistently outperforming traditional interpretation methods. Notably, the close performance between the stacking model ($R^2 = 0.9445$) and neural network ($R^2 = 0.9523$) suggests convergence near the uncertainty limits of ground truth measurements, highlighting the maturity of these techniques.

A key advantage lies in the direct prediction approach, which eliminates intermediate parameter estimation and reduces cumulative interpretation errors. This is further supported by SHAP analysis, which confirms resistivity as the dominant feature—aligning with Archie's law and reinforcing the model's interpretability and trustworthiness. The models demonstrated strong generalization across blind wells from different drilling campaigns, reservoir zones, and production states, validating their robustness in handling geological variability typically encountered in field deployments.

Beyond technical performance, the models offer significant operational and economic benefits. Automated interpretation reduces time and manpower requirements by over 90%, enabling rapid analysis of legacy wells and consistent field-wide saturation estimation. This not only helps uncover bypassed pay in older wells but also supports agile decision-making for infill drilling, workovers, and waterflood optimization. Moreover, results are objective, reproducible, and free from interpreter bias, improving input quality for reserves estimation and development planning.

To ensure successful deployment, emphasis must be placed on data quality control, outlier detection, and domain-informed feature engineering. Ensemble techniques should be leveraged for robustness, and models should be routinely validated against new data to monitor drift. Clear documentation of applicability limits is essential to guide appropriate usage and avoid overgeneralization.

Looking ahead, future efforts will focus on extending the framework to multi-well spatial modeling, time-lapse saturation tracking, and integration of physics-based constraints. Incorporating uncertainty quantification and transfer learning will further improve adaptability across fields. With these advancements, machine learning stands as a mature and practical tool—not just a promising technology—for enhancing petrophysical workflows and accelerating digital transformation in reservoir characterization.

Future Work

- Some S_w differences between predicted and actual are noted in the best part of the reservoir : 10350 ft – 10600 ft (Figure 15), this could be related to fluid type changes or lithology changes. A separate model will help us to get better results in this section, aimed as a part to progress this work further.
- Slight S_w differences between predicted and actual are noted in the high S_w values. This also will require a separate model to improve the prediction, aimed as a part to progress this work further.

- The observed high-frequency variations in blind well Sw predictions, particularly in heterogeneous intervals, suggest the need for depth-aware or sequence-based models (e.g., LSTM networks) that can capture stratigraphic context.

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The authors would like to express their sincere gratitude to Equinor for making the Volve field dataset publicly available, enabling open research and method development in reservoir characterization [4]. This comprehensive dataset has proved invaluable for advancing machine learning applications in petrophysics.

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Abbreviations

Well Logging Parameters

CALI	: Caliper log
DEN	: Bulk Density log
DTC	: Compressional sonic transit time
DTS	: Shear sonic transit time
GR	: Gamma Ray log
NEU	: Neutron Porosity log
PEF	: Photoelectric Factor
PHID	: Density-derived porosity
PHIF	: Total porosity
RDEP	: Deep Resistivity log
RMED	: Medium Resistivity log
ROP	: Rate of Penetration
VSH	: Volume of Shale

Petrophysical Parameters

a	: Archie's tortuosity factor
m	: Cementation exponent
n	: Saturation exponent
Rw	: Formation water resistivity
SCAL	: Special Core Analysis
Sw	: Water Saturation

Machine Learning Terms

CatBoost	: Categorical Boosting
FFNN	: Feedforward Neural Network
KNN	: K-Nearest Neighbors
L1/L2	: Lasso/Ridge regularization
LightGBM	: Light Gradient Boosting Machine
ML	: Machine Learning
MSE	: Mean Squared Error
PDDA	: Petrophysical Data-Driven Analytics
R²	: Coefficient of Determination
ReLU	: Rectified Linear Unit

RMSE : Root Mean Square Error
SHAP : SHapley Additive exPlanations
SVR : Support Vector Regression
XGBoost : eXtreme Gradient Boosting

Engineered Features

DEN_NEU_ratio : Density-Neutron Ratio
GR_norm : Normalized Gamma Ray
log_rd : Logarithmic Resistivity

Units of Measurement

API : American Petroleum Institute (gamma ray units)
ft : feet
g/cm³ : grams per cubic centimeter
ohm.m : ohm-meters
v/v : volume per volume (fraction)

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